

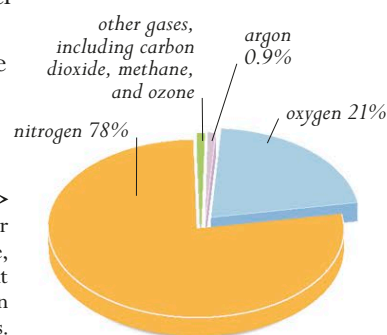
FRAGILE ATMOSPHERE

Earth's atmosphere is 120 km (75 miles) thick. It is made up of several layers, each with its own temperature and gas composition. Its density decreases with height, until it becomes the sparse, outermost layer, called the ionosphere. The ozone layer, in the lower atmosphere, plays a vital role in protecting life, because it absorbs harmful radiation such as ultraviolet light, which damages living cells. Before the ozone layer formed, life was confined to the seas, whose waters offered some protection against ultraviolet light.

The majority of water vapour and weather activity is restricted to the lowest 16 km (10 miles) of atmosphere, known as the troposphere. Earth's surface water and gaseous atmosphere interact to recycle water from the surface into the atmosphere and – through clouds, rain, and snow – redistribute it over the land and sea. From the land, water flows back into the sea, although large quantities are held back in lakes, ice, and under the ground.

ATMOSPHERIC GASES >

Nitrogen and oxygen make up over 99 per cent of Earth's atmosphere, along with small but important volumes of water vapour, carbon dioxide, and several other gases.



^ BLUE PLANET

Around two-thirds of Earth's surface is covered with water, which supports the abundance and diversity of life.



^ LAYERS OF ATMOSPHERE

Earth is surrounded by a thin, layered atmosphere, composed of water vapour and various gases, which trap solar energy and heat the surface.

VARIED ROCKS

There are around 500 different kinds of rocks on Earth, made up of varying combinations of thousands of naturally occurring minerals. All rocks have a specific composition and properties, and can be divided into three main categories: igneous rocks were originally molten; sedimentary rocks are deposited at Earth's surface; and metamorphic rocks result

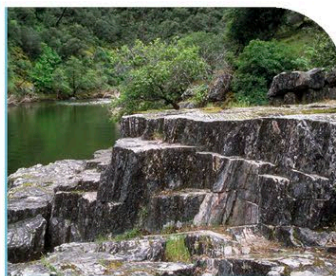
from the alteration of existing rocks within Earth's crust. These different types of rock are exposed on the surface by a mixture of uplift, driven by Earth's moving crust, and surface processes such as weathering and erosion. Erosion also modifies the rocks to produce multiple kinds of landforms, soils, and sediments. These are the inorganic elements on which life depends.



IGNEOUS ROCKS

The cooling and solidification of molten rock produces crystalline igneous rocks. Their composition and texture vary – rapid cooling produces fine grained rocks, and slow cooling produces coarse grained rocks.

BASALT



METAMORPHIC ROCKS

The application of heat and pressure to existing rocks deep within Earth's crust can change their form and mineral composition, resulting in metamorphic rocks such as slate, schist, and marble.

BIOTITE SCHIST



SEDIMENTARY ROCKS

Layers of sediment and the remains of plants and animals are deposited by wind and water. Over time, older and deeper buried sediments undergo compression and chemical change which transforms them into rock.

SANDSTONE

